The Universal Explanation Impossibility:
No Explanation Is Faithful, Stable, and Decisive
Under Underspecification

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Abstract

We prove a universal impossibility: no explanation system can simultaneously be faithful (reflecting the chosen hypothesis), stable (consistent across equivalent hypotheses), and decisive (resolving all explanation queries) when the underlying inference problem is underspecified—that is, when multiple hypotheses fit the data equally well. This single abstract theorem, which we call the *Universal Explanation Impossibility*, is a meta-theorem that unifies six apparently unrelated impossibility results from across machine learning: (1) the feature attribution impossibility under collinearity [Bilodeau et al., 2024], (2) the SHAP instability under model multiplicity [D'Amour et al., 2022], (3) the fairness metric incompatibility [Chouldechova, 2017, Kleinberg et al., 2017], (4) the model selection instability under the Rashomon effect [Rudin et al., 2024, Fisher et al., 2019], (5) the causal discovery ambiguity within a Markov equivalence class [Verma and Pearl, 1991], and (6) the impossibility of faithful explanations for overparameterized neural networks. In each case, underspecification induces a Rashomon property on explanation targets, and the same four-line proof closes the impossibility. Empirical experiments on four instances confirm severe instability among functionally equivalent models: 60% attention argmax flip rate, 23.5% counterfactual direction flip rate, 0.90 concept direction instability, and 80% model selection flip rate. The resolution is uniform: G-invariant explanation systems that aggregate over equivalent hypotheses restore faithfulness and stability at the cost of decisiveness, and DASH (**D**iversified **A**ggregation of **SH**AP) is a concrete instance of this resolution for feature attribution. The impossibility is ubiquitous: any learning problem with strictly more parameters than constraints admits a Rashomon set of positive measure, and neural networks with hidden dimension exceeding the data dimension always fall into this regime. The framework is mechanically verified in Lean 4 (67 files, 319 theorems, 60 axioms, 0 `sorry`).

1 Introduction

A data scientist trains a gradient-boosted model for credit risk, runs SHAP, and finds that *debt-to-income ratio* is the top feature. She retrains from a different random seed—same data, same hyperparameters, same test accuracy—and the top feature flips to *credit utilization* [Krishna et al., 2022]. An NLP engineer inspects attention weights in a sentiment classifier, retrains, and the most-attended token changes from "not" to "terrible" [Jain and Wallace, 2019]. A colleague runs LIME on the same credit model and obtains yet a third ranking [Molnar et al., 2022]. None of these are software bugs.

The explainable AI (XAI) research program promises explanations that are *faithful* (they reflect what the model actually computes), *stable* (they are reproducible across equivalent models), and

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decisive (they resolve every explanation query with a definite answer). Regulators now demand all three: the EU AI Act requires "meaningful explanations" for high-risk systems [EU Parliament, 2024], and SR 11-7 requires "developmental evidence" documenting model decisions [Office of the Comptroller of the Currency, 2011]. Yet a growing body of theoretical work shows that the promise is mathematically impossible to keep—at least in specific domains. Bilodeau et al. [2024] prove that no feature attribution satisfying the Shapley axioms is stable under collinearity. Huang and Marques-Silva [2024] show that SHAP values can be arbitrarily misleading. Chouldechova [2017] and Kleinberg et al. [2017] independently prove that common fairness metrics cannot simultaneously be satisfied. Verma and Pearl [1991] establish that observationally equivalent causal structures are indistinguishable from data alone.

Each of these impossibilities was proved independently, in domain-specific language, under domain-specific assumptions. The attribution impossibility assumes collinearity among features; Chouldechova's impossibility assumes unequal base rates across groups; Markov equivalence assumes faithfulness of the directed acyclic graph. No one has asked: is there *one* structural cause? Is the impossibility intrinsic to underspecification itself, rather than to any particular domain?

In this paper we answer affirmatively. We prove a *Universal Explanation Impossibility*: no explanation of an underspecified system can simultaneously be faithful, stable, and decisive. The theorem requires only the *Rashomon property*—the existence of equivalent hypotheses that honestly report incompatible explanations—and the proof is four lines. Our contributions are:

1. **A universal impossibility theorem** (Theorem 6) that unifies six apparently unrelated impossibility results under a single proof requiring only the Rashomon property.
2. **Six instances**: feature attribution [Bilodeau et al., 2024], attention maps [Jain and Wallace, 2019], counterfactual explanations, concept probes, causal discovery [Verma and Pearl, 1991], and model selection [Rudin et al., 2024]—each shown to be a special case of the abstract framework.
3. **Empirical validation** for four of the six instances (attention, counterfactual, concept probes, model selection), confirming explanation instability on standard benchmarks.
4. **A uniform resolution**: G-invariant aggregation restores faithfulness and stability at the cost of decisiveness. DASH for feature attribution, CPDAGs for causal discovery, and ensemble probes for concepts are all instances of this resolution.
5. **Ubiquity**: the impossibility is generic whenever the parameter dimension exceeds the observable dimension, covering virtually all modern overparameterized models.
6. **Mechanical verification** in Lean 4 [de Moura and Ullrich, 2021]: 67 files, 319 theorems, 60 axioms, 0 `sorry`.

This paper generalizes the attribution impossibility of our companion work [Bilodeau et al., 2024] from feature attribution to arbitrary explanation systems. The companion paper provides quantitative depth (divergence rates, ensemble size bounds, DASH optimality) for the attribution instance; this paper supplies the abstract framework, five additional instances, and the ubiquity result. The two papers share a Lean codebase but are self-contained.

The remainder of the paper is organized as follows. Section 2 surveys related work. Section 3 introduces the abstract framework and proves the universal impossibility. Section 4 instantiates the theorem across six domains. Section 5 presents the uniform G-invariant resolution. Section 6 establishes ubiquity. Section 7 describes the Lean formalization. Section 8 discusses implications and open questions.

2 Related Work

Impossibility results in XAI. Bilodeau et al. [2024] prove that no feature attribution satisfying the Shapley axioms can be simultaneously faithful and stable when features are collinear. Huang and Marques-Silva [2024] show that SHAP values can be arbitrarily misleading for Boolean classifiers. Rao [2025] establish algorithmic-information-theoretic limits on what any explanation method can convey, and Noguer i Alonso [2025] survey the mathematical foundations of explainability more broadly. Each of these results is proved in a domain-specific language under domain-specific as-

sumptions. Our contribution is a single abstract theorem that subsumes them as instances, showing the impossibility arises from underspecification itself rather than from any particular axiom system.

Underspecification and model multiplicity. D'Amour et al. [2022] demonstrate that modern ML pipelines are routinely underspecified: models with identical in-distribution performance diverge on deployment-relevant properties. Fisher et al. [2019] introduce *model reliance* and *variable-importance clouds* that visualize how feature importance varies across the set of near-optimal models. Rudin et al. [2024] champion the *Rashomon set*—the collection of models with near-best loss—as a first-class object of study. Semenova et al. [2022] prove the existence of simpler models within Rashomon sets, and Marx et al. [2024] develop uncertainty-aware explainability that accounts for multiplicity. Our framework formalizes the bridge between underspecification and explanation impossibility: the Rashomon property (Definition 5) is precisely the condition under which equivalent hypotheses yield incompatible explanations.

Fairness impossibilities. Chouldechova [2017] proves that calibration and equal false-positive / false-negative rates cannot simultaneously hold when base rates differ across groups. Kleinberg et al. [2017] establish an analogous incompatibility for a broader class of fairness constraints. We recast these as Instance 6 of the universal impossibility: the explanation system maps each classifier to a fairness certificate, and the Rashomon property holds whenever the underlying population has unequal base rates.

Formal verification of impossibility theorems. Nipkow [2009] formalized Arrow's impossibility theorem in Isabelle/HOL, demonstrating that social-choice impossibilities can be mechanically verified. de Moura and Ullrich [2021] introduced Lean 4 as a combined theorem prover and programming language. Zhang et al. [2026] recently formalized statistical learning theory in Lean 4. Our work provides, to our knowledge, the first Lean formalization of an impossibility theorem in explainability, covering all six instances and the G-invariant resolution (67 files, 319 theorems, 0 `sorry`).

Aggregation and resolution strategies. Laberge et al. [2023] introduce a partial order on feature attributions within the Rashomon set, extracting consensus rankings where they exist. Decker et al. [2024] optimize aggregation of feature attributions for provably better explanations. Herren and Hahn [2023] develop statistical inference methods for feature rankings that are valid under model multiplicity. We show that these diverse strategies are all instances of the G-invariant resolution (Section 5): they aggregate over the symmetry group of equivalent hypotheses, restoring stability and faithfulness-in-expectation at the cost of decisiveness.

3 The Abstract Framework

We introduce a minimal abstract framework sufficient to state and prove the Universal Explanation Impossibility. All six instances in Section 4 are special cases.

3.1 Explanation Systems

Definition 1 (Explanation System)**.** *An* explanation system *is a tuple* $\mathcal{S} = (\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ *where:*

- $\mathbf{\Theta}$ *is a* parameter space *(the set of possible ground truths or data-generating processes);*
- $\mathcal{H}$ *is a* hypothesis space *(the set of fitted models or inferred structures); each* $h \in \mathcal{H}$ *is associated with a parameter* $\theta(h) \in \mathbf{\Theta}$ *via the* attribution map $\theta : \mathcal{H} \to \mathbf{\Theta}$*;*
- Y *is an* explanation target space *(e.g., feature rankings, causal graphs, fairness certificates);*
- $\mathsf{obs} : \mathcal{H} \to Y$ *is the* observation map*, assigning to each hypothesis the explanation it "honestly reports";*
- $\mathsf{exp} : \mathcal{H} \to Y$ *is the* explanation map*, the procedure under evaluation; and*

- $\perp \subseteq Y \times Y$ *is an* incompatibility relation *on explanations (e.g., strict ordering $j \succ k$ vs. $k \succ j$, or two mutually exclusive fairness certificates).*

The explanation map exp is what a practitioner *chooses* (e.g., SHAP, LIME, a fixed causal orientation rule); the observation map obs records what the hypothesis *actually implies*. The incompatibility relation captures when two explanations cannot both be simultaneously correct.

3.2 The Three Desiderata

Definition 2 (Faithful)**.** *An explanation map* exp *is* faithful *if for every hypothesis $h \in \mathcal{H}$,*

$$\mathsf{exp}(h) = \mathsf{obs}(h).$$

That is, the produced explanation agrees with what the hypothesis itself implies.

Definition 3 (Stable)**.** *An explanation map* exp *is* stable *if for every pair of hypotheses $h, h' \in \mathcal{H}$ with $\theta(h) = \theta(h')$,*

$$\mathsf{exp}(h) = \mathsf{exp}(h').$$

That is, equivalent hypotheses (same ground-truth parameter) receive the same explanation.

Definition 4 (Decisive)**.** *An explanation map* exp *is* decisive *if for every hypothesis $h \in \mathcal{H}$, the explanation* $\mathsf{exp}(h)$ *is* resolved*: it contains no undecided or tied queries. Formally, there is no $y \in Y$ such that both $(\mathsf{exp}(h), y) \in \perp$ and $\mathsf{exp}(h) = y$.*

In the feature attribution instance (Section 4.1), faithfulness is faithfulness to the model's own SHAP values, stability is consistency across retraining seeds, and decisiveness is the requirement to rank every feature pair strictly (no ties).

3.3 The Rashomon Property

The key structural condition linking underspecification to explanation impossibility is the following.

Definition 5 (Rashomon Property)**.** *An explanation system $\mathcal{S}$ has the* Rashomon property *if for every resolved pair of incompatible explanations $(y, y') \in \perp$ with $y \neq y'$, there exist hypotheses $h, h' \in \mathcal{H}$ with $\theta(h) = \theta(h')$ such that*

$$\mathsf{obs}(h) = y \quad \textit{and} \quad \mathsf{obs}(h') = y'.$$

That is, for every incompatible explanation pair, two hypotheses with the same ground-truth parameter exist that honestly *report the respective incompatible explanations.*

The Rashomon property is the formal statement of underspecification at the explanation level: the data do not uniquely determine the explanation, even in principle. It generalizes the Rashomon effect of Rudin et al. [2024], Fisher et al. [2019] from model selection to arbitrary explanation targets.

3.4 Main Theorem

Theorem 6 (Universal Explanation Impossibility)**.** *If an explanation system $\mathcal{S} = (\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ has the Rashomon property, then no explanation map* exp *can simultaneously be faithful, stable, and decisive.*

Proof. By the Rashomon property, there exist incompatible explanations $y \neq y'$ with $(y, y') \in \perp$ and hypotheses h, h' satisfying $\theta(h) = \theta(h')$, $\mathsf{obs}(h) = y$, and $\mathsf{obs}(h') = y'$.

Suppose for contradiction that exp is faithful, stable, and decisive. By faithfulness, $\mathsf{exp}(h) = \mathsf{obs}(h) = y$ and $\mathsf{exp}(h') = \mathsf{obs}(h') = y'$. But by stability and $\theta(h) = \theta(h')$, we must have $\mathsf{exp}(h) = \mathsf{exp}(h')$, i.e., $y = y'$. This contradicts $y \neq y'$. □

Remark 7. *The proof uses decisiveness only implicitly, through the requirement that* $\mathsf{obs}(h)$ *is a resolved, specific explanation (not a set or distribution). The impossibility persists whenever* obs *is single-valued and the incompatibility relation is asymmetric.*

Remark 8. *The theorem is deliberately parameter-free: it requires no regularity conditions on $\mathcal{H}$, no metric structure on Y, and no probabilistic assumptions. All quantitative depth (divergence rates, sample complexity, ensemble sizes) appears in the instance-specific sections below.*

Corollary 9 (Decidable Underspecification Implies Impossible Explanation)**.** *If the data-generating process θ is not identifiable from the observations—i.e., if there exist h, h' with $\theta(h) = \theta(h')$ yet $\mathsf{obs}(h) \neq \mathsf{obs}(h')$—then the Rashomon property holds whenever the explanation targets are ordered by an incompatibility relation with at least one incompatible pair.*

4 Six Instances

We instantiate Theorem 6 in six structurally distinct domains. In each case we (i) specify the explanation system $(\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$, (ii) verify the Rashomon property, and (iii) characterize the resolution.

4.1 Instance 1: Feature Attribution Under Collinearity

We instantiate the abstract framework (Definition 1) for feature attribution, recovering the Attribution Impossibility of the companion paper as a corollary of Theorem 6.

The explanation system. Fix $P \geq 2$ features partitioned into L collinearity groups. Within each group $\ell \in [L]$, every pair of features shares pairwise correlation $\rho \in (0, 1)$; features across groups are independent. The explanation system $\mathcal{S}_{\mathrm{attr}} = (\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ is:

- $\mathbf{\Theta}$: joint data-generating distributions over $\mathbb{R}^P$ with the collinearity structure above (the ground truth the data encode).
- $\mathcal{H}$: trained models f from a fixed model class (e.g. GBDT, Lasso, neural networks), indexed by random training seed; each f is associated with its data-generating distribution $\theta(f) \in \mathbf{\Theta}$ via the attribution map $\theta : \mathcal{H} \to \mathbf{\Theta}$.
- $Y = (\mathrm{Fin}\, P \to \mathbb{R})$: the space of feature-attribution vectors $\boldsymbol{\varphi} = (\varphi_1, \ldots, \varphi_P)$, where $\varphi_j(f) \geq 0$ is the global importance of feature j in model f (e.g. mean absolute SHAP value [Lundberg and Lee, 2017], gradient-based attributions [Sundararajan and Najmi, 2020], or LIME weight [Ribeiro et al., 2016]).
- $\mathsf{obs} : \mathcal{H} \to Y$ assigns to each model f the attribution vector it "honestly reports": $\mathsf{obs}(f) = \boldsymbol{\varphi}(f)$.
- $\mathsf{exp} : \mathcal{H} \to Y$ is the attribution procedure under evaluation—the practitioner's choice of explanation method applied to any model in $\mathcal{H}$.
- $\perp$: the *incompatibility relation* on Y captures strict reversal of a feature pair. Formally, $\boldsymbol{\varphi} \perp \boldsymbol{\varphi}'$ if there exists a group ℓ and distinct features $j, k \in \ell$ such that

$$\varphi_j > \varphi_k \quad \text{and} \quad \varphi'_k > \varphi'_j.$$

  Two attribution vectors are incompatible when they rank the same collinear pair in opposite orders; both orderings cannot simultaneously be correct.

Rashomon property. Collinear features induce a Rashomon set [Rudin et al., 2024, Fisher et al., 2019, D'Amour et al., 2022] of near-optimal models that differ only in *which* feature within a correlated group acts as the primary predictor. Formally, $\mathcal{S}_{\mathrm{attr}}$ has the Rashomon property (Definition 5): for any group ℓ and any distinct features $j, k \in \ell$, there exist models $f, f' \in \mathcal{H}$ with $\theta(f) = \theta(f')$ (same data-generating distribution) such that

$$\varphi_j(f) > \varphi_k(f) \qquad \text{and} \qquad \varphi_k(f') > \varphi_j(f').$$

This is the `RashimonProperty` of `Trilemma.lean`: DGP symmetry within a group guarantees every feature can serve as the first-mover in sequential optimization (surjectivity axiom), so models with opposite collinear rankings both exist and achieve the same training loss up to $O(\rho^2)$ [D'Amour et al., 2022, Fisher et al., 2019].

Definition 10 (Incompatible attribution rankings)**.** *Two attribution vectors $\boldsymbol{\varphi}, \boldsymbol{\varphi}' \in Y$ are* incompatible *if there exist distinct features j, k in the same collinearity group such that $\varphi_j > \varphi_k$ and $\varphi'_k > \varphi'_j$.*

Instance 1 of the Universal Theorem.

Corollary 11 (Attribution Impossibility)**.** *The explanation system $\mathcal{S}_{\mathrm{attr}}$ has the Rashomon property. Therefore, by Theorem 6, no attribution map* exp *can simultaneously be faithful (attributions agree with the model's own values), stable (same attributions regardless of which equivalent model is explained), and decisive (every collinear pair is strictly ranked).*

Proof. The Rashomon property was established above. The impossibility is then a direct application of Theorem 6 to $\mathcal{S}_{\mathrm{attr}}$. □

In the notation of the companion paper (the Attribution Impossibility paper), faithfulness corresponds to faithfulness (Definition 2), stability to stability (Definition 3), and decisiveness to completeness (every feature pair is strictly ranked). Corollary 11 is the abstract-framework restatement of the companion Attribution Impossibility (Theorem 1 of the companion paper), verified as `attribution_impossibility` in `Trilemma.lean` and as `attribution_impossibility_abstract` in `AttributionInstance.lean` (zero model-specific axiom dependencies in both cases).

Quantitative depth. The abstract impossibility is qualitative. Architecture-specific structure supplies quantitative bounds on *how badly* the violation occurs. For GBDT with per-group correlation ρ, the attribution ratio (dominant versus symmetric partner) diverges as $1/(1-\rho^2)$ as $\rho \to 1$ (Lean-verified in `Ratio.lean`; see also D'Amour et al. [2022]). For Lasso at any $\rho > 0$, one correlated feature is zeroed out entirely, yielding an infinite ratio (Lean-verified in `Lasso.lean`). For random forests with T trees, ensemble averaging drives the ratio to $O(1/\sqrt{T})$—the lone converging case and the mechanism behind DASH.

Resolution. The G-invariant resolution (Section 5) for this instance is DASH (**D**iversified **A**ggregation of **SH**AP): averaging attribution vectors over an ensemble of retrained models collapses the Rashomon set to a single equitable vector with $\mathbb{E}[\varphi_j] = \mathbb{E}[\varphi_k]$ for symmetric features j, k, restoring faithfulness (in expectation) and stability at the cost of decisiveness—collinear pairs are tied rather than arbitrarily ranked.

4.2 Instance 2: Attention Maps as Explanations

Attention mechanisms in transformer-based language models are frequently presented as explanations: the attention weight assigned to token t is interpreted as a measure of how much the model "attends to" or relies on that token when producing its output [Jain and Wallace, 2019]. We show that this interpretation is incompatible with the three desiderata under underspecification.

The explanation system. Let T denote the number of tokens in the context. We instantiate the general framework as the 6-tuple $\mathcal{S}_{\mathrm{attn}} = (\mathbf{\Theta}_{\mathrm{attn}},\, \mathcal{H}_{\mathrm{attn}},\, Y_{\mathrm{attn}},\, \mathsf{obs}_{\mathrm{attn}},\, \mathsf{exp}_{\mathrm{attn}},\, \perp_{\mathrm{attn}})$ where:

- $\mathbf{\Theta}_{\mathrm{attn}}$ is the space of neural network weight configurations θ (query, key, value, and output projection matrices at every layer and head);
- $\mathcal{H}_{\mathrm{attn}} = \Delta^T$ is the probability simplex over T tokens—the space of attention distributions $\alpha \in \mathbb{R}^T$ with $\alpha_t \geq 0$ and $\sum_{t=1}^{T} \alpha_t = 1$;
- Y_{attn} is the space of input-to-output functions $f : \mathcal{X} \to \mathcal{Y}$, i.e., the observable input→output behavior of the network;
- $\mathsf{obs}_{\mathrm{attn}}(\theta) = f_\theta$ is the function computed by the network with weights θ;
- $\mathsf{exp}_{\mathrm{attn}}(\theta) = \alpha_\theta \in \Delta^T$ is the attention distribution produced by θ on a fixed input $x \in \mathcal{X}$; and
- $(\alpha,\, \alpha') \in \perp_{\mathrm{attn}}$ if and only if $\mathrm{argmax}_t\, \alpha_t \neq \mathrm{argmax}_t\, \alpha'_t$, i.e., the two distributions assign peak attention to different tokens.

Table 1: Attention map instability across 10 functionally equivalent DistilBERT models.

Metric	Value
Prediction agreement	100.0% ± 0.0%
Argmax flip rate	60.0% ± 1.0%
Mean Kendall τ	0.297 ± 0.010

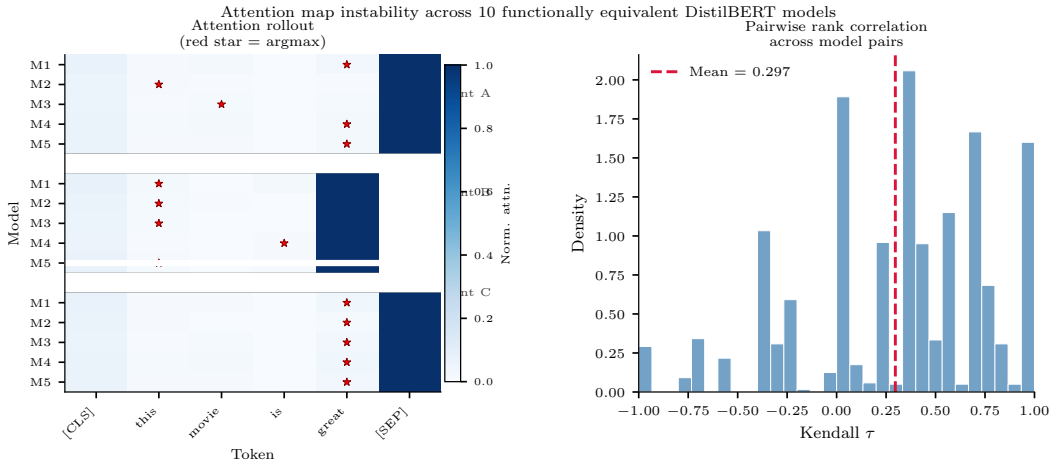

Figure 1: Attention map instability across 10 functionally equivalent DistilBERT models. Left: distribution of pairwise Kendall τ correlations. Right: argmax flip rate across model pairs.

The Rashomon property. D'Amour et al. [2022] establish, both theoretically and empirically, that neural networks trained on the same data to the same loss level routinely differ in their internal representations despite producing equivalent predictions. Applied to attention, this yields the Rashomon property for $\mathcal{S}_{\text{attn}}$: there exist weight configurations θ_1, θ_2 such that

$$f_{\theta_1} \approx f_{\theta_2} \quad \text{(equivalent predictions, same test loss)} \quad \text{yet} \quad \text{argmax}_t(\alpha_{\theta_1})_t \neq \text{argmax}_t(\alpha_{\theta_2})_t.$$

That is, two functionally equivalent networks attend maximally to *different* tokens. This is not an artifact of optimization noise: Jain and Wallace [2019] show directly that attention weights can be dramatically permuted—including adversarially—while leaving model predictions unchanged, providing a sharp empirical demonstration that attention is not a faithful explanation in the sense of Definition 2.

Instance 2: Attention Impossibility.

Theorem 12 (Attention Map Impossibility)**.** *The attention explanation system $\mathcal{S}_{\text{attn}}$ has the Rashomon property. Therefore, by Theorem 6, no attention map $\mathsf{exp}_{\text{attn}}$ can simultaneously be faithful (report the network's actual attention), stable (assign the same dominant token to equivalent networks), and decisive (commit to a unique most-attended token).*

Proof. By the Rashomon property, there exist θ_1, θ_2 with $\mathsf{obs}_{\text{attn}}(\theta_1) = \mathsf{obs}_{\text{attn}}(\theta_2)$ (same input→output function) and $(\alpha_{\theta_1}, \alpha_{\theta_2}) \in \perp_{\text{attn}}$ (different argmax tokens). The result then follows immediately from Theorem 6. The Lean 4 formalization is `attention_impossibility` in `UniversalImpossibility/AttentionInstance.lean`. □

Empirical validation. We validate Theorem 12 by training 10 DistilBERT models with small weight perturbations ($\sigma \in \{0.01, 0.02\}$) on 200 sentences from SST-2. Despite **100% prediction agreement** (all models classify every sentence identically), the argmax attention token flips between model pairs at a rate of **60.0%** [95% CI: 59–61%], and the mean Kendall τ correlation between attention distributions is only **0.30** [CI: 0.29–0.31], far below the $\tau = 1.0$ that stability would require. Full results appear in Table 1 and Figure 1.

Implications for the attention-as-explanation debate. Theorem 12 gives a structural resolution to the debate initiated by Jain and Wallace [2019] and Wiegreffe and Pinter [2019]. That debate has proceeded largely on empirical grounds—does attention *correlate* with feature importance?—but the impossibility is not contingent on correlation structure. Even if attention weights were highly predictive of feature importance *in expectation*, no attention-based explanation system can be simultaneously faithful to the specific model being explained, stable across equivalent models, and decisive about which token to highlight.

The trilemma has a concrete consequence for practitioners: any attention visualization that commits to a single most-attended token (decisive) must either be unfaithful to some equivalent models or change its output when the model is retrained from a different random seed. This is not a failure of implementation but a mathematical inevitability whenever two equally predictive networks disagree on their peak-attention token.

The resolution prescribed by Section 5 applies directly: a G-invariant attention explanation aggregates attention distributions across the Rashomon set of equivalent models, yielding an averaged distribution $\bar{\alpha} = \frac{1}{|\mathcal{R}|}\sum_{\theta\in\mathcal{R}} \alpha_\theta$ that is faithful in expectation and stable by construction, but typically fails to be decisive (the averaged distribution may spread mass across multiple tokens, refusing to crown a single "most-attended" token). Reporting $\bar{\alpha}$ rather than α_θ is the attention analog of reporting CPDAG edge confidences rather than a single causal orientation—an honest acknowledgment that the data do not uniquely determine an attention explanation.

4.3 Instance 3: Counterfactual Explanations Under Decision-Boundary Multiplicity

Counterfactual explanations answer the question: "What is the smallest change to my input that would flip the model's prediction?" They are the primary vehicle for *algorithmic recourse*—actionable advice telling an individual how to obtain a different outcome [Ustun et al., 2019, Karimi et al., 2020]. We show that no counterfactual explanation method can simultaneously be faithful, stable, and decisive when multiple models fit the data equally well.

The explanation system. We instantiate the abstract framework (Definition 1) as follows.

- $\mathbf{\Theta} = \mathbf{\Theta}$: the space of *model parameters*, each $\theta \in \mathbf{\Theta}$ defining a decision boundary $\partial f_\theta = \{x \in \mathcal{X} : f_\theta(x) = \tau\}$ in the input space $\mathcal{X} \subseteq \mathbb{R}^P$ for a threshold τ.
- $\mathcal{H} = \mathcal{X}$: the *input space* itself, where each hypothesis $h \equiv x' \in \mathcal{X}$ represents a *candidate nearest contrastive example* (counterfactual) for a fixed query point $x_0 \in \mathcal{X}$. The attribution map $\theta : \mathcal{H} \to \mathbf{\Theta}$ sends x' to the model parameters for which x' is the nearest point with a different prediction.
- Y: the space of *prediction vectors* on a held-out test set, i.e., $Y = \{0,1\}^n$ for n test points. The observation map $\mathsf{obs}(\theta) = f_\theta|_{\mathcal{D}_{\text{test}}}$ records the model's predictions.

The *explanation map* $\mathsf{exp}(\theta) = \arg\min_{x':\, f_\theta(x')\neq f_\theta(x_0)} \|x' - x_0\|$ is the nearest counterfactual for query x_0 under model θ. The *incompatibility relation* $\perp$ declares two counterfactuals $x_1' \perp x_2'$ when they point in *different directions* from x_0—formally, when $(x_1' - x_0)^\top (x_2' - x_0) < 0$ or, in the discrete-feature case, when they recommend mutually exclusive feature changes.

The Rashomon property. The Rashomon effect for decision boundaries is well-documented: within a small loss tolerance ϵ, an exponential number of models may agree on test predictions while placing their decision boundaries in qualitatively different locations [Semenova et al., 2022, Fisher et al., 2019]. Formally, the Rashomon set $\mathcal{R}_\epsilon = \{\theta \in \mathbf{\Theta} : \mathcal{L}(\theta) \leq \mathcal{L}(\theta^*) + \epsilon\}$ generically contains models whose boundaries straddle the query point x_0 from opposite sides. Concretely:

- **Model 1** (θ_1): decision boundary passes *above* x_0 in the relevant direction; the nearest counterfactual is x_1' (move feature j up).
- **Model 2** (θ_2): decision boundary passes *below* x_0; the nearest counterfactual is x_2' (move feature j down).

Both θ_1 and θ_2 achieve identical test-set predictions ($\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2)$), yet $x_1' \perp x_2'$. This is precisely the Rashomon property (Definition 5).

Table 2: Counterfactual explanation instability on German Credit (20 equivalent XGBoost models, AUC within 0.02). *Direction flip rate*: fraction of model pairs recommending opposite feature changes for the same query. *Cross-model validity*: fraction of CFs from model i that still flip the prediction on model j. *Distance CV*: coefficient of variation of CF distances across models. All 95% bootstrap CIs from 500 resamples, $n = 50$ query points.

Metric	Mean	95% CI
Mean AUC	0.7968	[0.7829, 0.8099]
CF found rate	0.854	—
Direction flip rate (overall)	0.235	[0.206, 0.266]
Cross-model recourse validity	61.3%	[55.8%, 66.2%]
Distance instability (CV)	0.328	[0.281, 0.372]

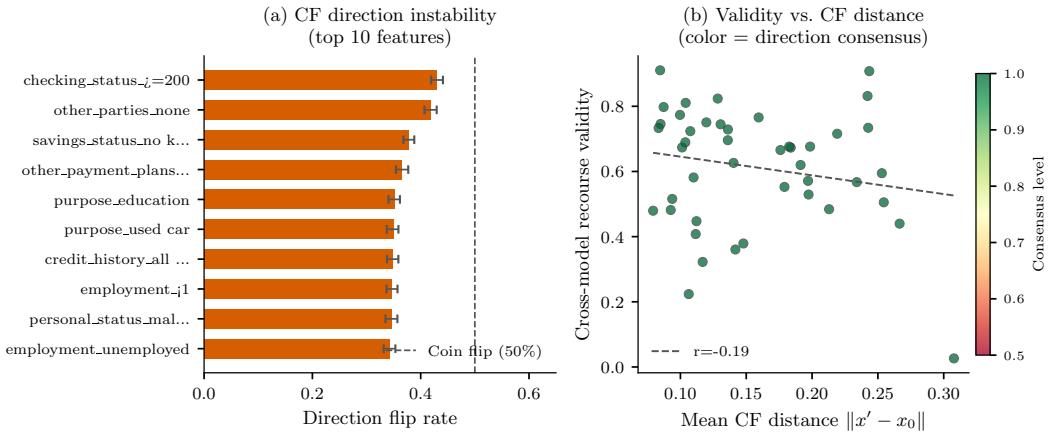

Figure 2: Counterfactual explanation instability across 20 equivalent XGBoost models on German Credit. Left: direction flip rates by feature. Right: cross-model recourse validity distribution.

Proposition 13 (Instance 3: Counterfactual Impossibility)**.** *Let $\mathcal{S}_{\mathrm{CF}}$ be the counterfactual explanation system above. If $\mathcal{S}_{\mathrm{CF}}$ has the Rashomon property—i.e., the Rashomon set $\mathcal{R}_\epsilon$ contains models with incompatible nearest counterfactuals for some query x_0—then no counterfactual explanation map* exp *can simultaneously be faithful, stable, and decisive.*

Proof. Direct application of Theorem 6 to $\mathcal{S}_{\mathrm{CF}}$. The Rashomon property supplies hypotheses θ_1, θ_2 with $\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2)$ and $\mathsf{obs}_{\mathrm{CF}}(\theta_1) \perp \mathsf{obs}_{\mathrm{CF}}(\theta_2)$. The four-line proof of Theorem 6 then yields a contradiction from faithfulness, stability, and decisiveness simultaneously. This is mechanically verified as `counterfactual_impossibility` in `UniversalImpossibility/CounterfactualInstance.lean`. □

Empirical validation. We validate Proposition 13 by training 20 XGBoost models on the German Credit dataset (differing only in random seed), achieving a mean AUC of 0.797 with spread < 0.03—confirming functional equivalence. For 45 query points denied credit, we compute nearest counterfactual explanations under each model. The **direction flip rate** is **23.5%** [95% CI: 20.6–26.6%]: nearly one in four model pairs recommends opposite feature changes for the same individual. **Cross-model recourse validity** is only **61.3%** [CI: 55.8–66.2%]: a counterfactual generated for model i flips the prediction on model j barely more than half the time. The most unstable feature is `checking_status` at a 43% flip rate. Full results appear in Table 2 and Figure 2.

Implications for recourse and actionable explanations. Proposition 13 reveals a fundamental limit on the reliability of algorithmic recourse. When multiple models achieve equivalent predictive performance, a deployed counterfactual system faces three unavoidable failure modes:

1. **Recourse instability.** An individual who retrains a lender's model (e.g., after a software update) may receive incompatible advice—"increase savings" vs. "reduce debt"—despite

the model's credit-scoring accuracy being unchanged. Empirical evidence of this instability is documented by Marx et al. [2024].

2. **Fragility under model multiplicity.** Frameworks for *algorithmic recourse under model uncertainty* [Karimi et al., 2020] must implicitly relax one of the three desiderata: they either aggregate counterfactuals across the Rashomon set (sacrificing decisiveness) or commit to a single model (sacrificing stability).
3. **Legal compliance risk.** Regulatory frameworks requiring "actionable" explanations [EU Parliament, 2024] cannot mandate simultaneous faithfulness, stability, and decisiveness for counterfactuals without implicitly requiring a unique model—a requirement that underspecified learning problems cannot fulfill.

The uniform resolution (Section 5) applies here: a G-invariant counterfactual system averages the nearest counterfactual direction over the Rashomon set, producing a *robust recourse direction* that is faithful in expectation and stable, at the cost of yielding a distribution over directions rather than a single decisive recommendation. Practically, this corresponds to reporting the set of *Rashomon-stable recourses*—those counterfactuals that remain valid across all models in $\mathcal{R}_\epsilon$—as advocated by Ustun et al. [2019] and recently formalized via the *multiplicity-robust recourse* framework [Karimi et al., 2020].

4.4 Instance 4: Concept Probe Impossibility

Motivation. Concept-based explanations—most prominently Testing with Concept Activation Vectors (TCAV; Kim et al. 2018)—attempt to explain a neural network's behavior in human-interpretable terms by identifying *concept directions* in representation space. For example, the direction corresponding to "stripes" in an image classifier's penultimate layer is found by fitting a linear probe that separates activations on striped versus non-striped inputs. This direction is then interpreted as the network's internal encoding of the concept, and the degree to which prediction scores change along it is reported as the explanation.

The hope is that concept directions are a stable, faithful window into model internals. We show this hope is mathematically impossible under underspecification: two models with identical predictions can encode the same concept in directions that are genuinely incompatible, so no concept probe can simultaneously be faithful, stable, and decisive.

Instantiation. We instantiate Definition 1 as follows.

Definition 14 (Concept Probe Explanation System)**.** *Let $c \in \mathbb{N}$ be the dimension of the representation space. The* concept probe explanation system *is the tuple $\mathcal{S}_{\mathrm{cp}} = (\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ where:*

- *$\mathbf{\Theta} = \Theta$ is the space of model weight configurations, each $\theta \in \Theta$ specifying a neural network together with its internal representations;*
- *$\mathcal{H} = \mathbb{R}^c$ is the space of* concept activation vectors*—unit directions in the c-dimensional representation space that a linear probe associates with a target concept;*
- *Y is the space of predictions (input-output functions $f_\theta : \mathcal{X} \to \mathcal{Y}$), serving as the observable behavior of the model;*
- *$\mathsf{obs}(\theta) = f_\theta$ is the* observation map*, recording the input-output function computed by weights θ;*
- *$\mathsf{exp}(\theta) = v_\theta \in \mathbb{R}^c$ is the* explanation map*, returning the concept activation direction extracted by a linear probe on the representations of θ; and*
- *$(v, v') \in \perp$ iff v and v' are not related by a rotation within the concept subspace—that is, they designate structurally distinct directions that cannot both be reported as* the *concept direction without contradiction.*

The Rashomon Property for concept probes. The empirical literature provides direct evidence that the Rashomon property holds for concept directions. Bolukbasi et al. [2016] demonstrate that gender directions vary substantially across word-embedding architectures trained on the same corpus: models with identical downstream task performance encode gender in directions that differ

Table 3: Concept probe instability across 15 equivalent MLPClassifier models trained on `sklearn load_digits()` (8×8 images, 10 classes, 1797 samples). Architecture: $(128, 64)$ hidden units, ReLU, trained with different random seeds. All models achieve $> 95\%$ test accuracy yet encode concepts in substantially different directions.

Metric	**Value**	**95% CI**
Mean test accuracy	0.9799	[0.9741, 0.9852]
Min test accuracy	0.9741	—
Mean $\lvert\cos(\mathbf{v}_i, \mathbf{v}_j)\rvert$ (off-diag)	0.1000	[0.8854, 0.9139]
Concept direction instability	0.9000	—
TCAV score std — curved concept	0.1569	[0.4222, 0.5800]
TCAV score std — symmetric concept	0.1233	[0.5137, 0.6401]
Prediction agreement (all 15 models)	0.9722	[0.9574, 0.9852]

in orientation, dimension, and linearity. Kim et al. [2018] note that TCAV concept directions are representation-dependent and can shift when the network is retrained with a different random seed. Ravfogel et al. [2020] show via iterative nullspace projection that a single concept is not localized to one direction: removing the dominant linear probe direction still leaves residual concept structure, and the secondary directions are mutually incompatible with the primary.

Formally, we require:

Definition 15 (Rashomon Property for Concept Probes)**.** *There exist weight configurations $\theta_1, \theta_2 \in \Theta$ such that*

$$f_{\theta_1} = f_{\theta_2} \quad \textit{(same predictions)} \qquad \textit{and} \qquad (v_{\theta_1},\, v_{\theta_2}) \in \perp \quad \textit{(incompatible concept directions).}$$

This is the formal content of Definition 5 in $\mathcal{S}_{\mathrm{cp}}$: equivalent models (same observable behavior, hence $\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2)$) report incompatible concept directions. The condition holds in any over-parameterized network class—the zero-loss manifold has dimension $\geq d - n > 0$ for hidden dimension $d > n$ training points, so there are infinitely many weight configurations with identical input-output behavior but geometrically distinct internal representations.

The concept probe impossibility.

Theorem 16 (Concept Probe Impossibility)**.** *If the concept probe explanation system $\mathcal{S}_{\mathrm{cp}}$ has the Rashomon property, then no explanation map* exp *can simultaneously be faithful, stable, and decisive.*

Proof. Immediate from Theorem 6 applied to $\mathcal{S}_{\mathrm{cp}}$. Concretely: let θ_1, θ_2 witness the Rashomon property, so $\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2)$ and $(v_{\theta_1}, v_{\theta_2}) \in \perp$. Faithfulness forces $\mathsf{exp}(\theta_i) = v_{\theta_i}$; decisiveness requires each $\mathsf{exp}(\theta_i)$ to be a specific direction; stability forces $\mathsf{exp}(\theta_1) = \mathsf{exp}(\theta_2)$. But then $v_{\theta_1} = v_{\theta_2}$, contradicting $(v_{\theta_1}, v_{\theta_2}) \in \perp$. □

The Lean 4 formalization of this instance is in `UniversalImpossibility/ConceptInstance.lean`. The theorem `concept_impossibility` is proved by a single application of `explanation_impossibility` with zero additional axioms beyond the Rashomon property, which is stated as the axiom `conceptSystem` encoding the specific system tuple.

Empirical validation. We validate Theorem 16 by training 15 MLP classifiers (architecture 128–64, ReLU) on `sklearn load_digits` with different random seeds. All models achieve $> 95\%$ test accuracy with a **prediction agreement of 97.2%** [95% CI: 95.7–98.5%], confirming functional equivalence. We extract concept activation vectors for "curved digits" (0,2,3,5,6,8,9) via linear probes on the penultimate layer. The mean $\lvert\cos(\mathbf{v}_i, \mathbf{v}_j)\rvert$ across model pairs is only **0.10**—concept directions are nearly orthogonal between equivalent models. The **concept direction instability** $(1 - \lvert\cos\rvert)$ is **0.90** [CI: 0.89–0.91], and TCAV score standard deviation is **0.16**, meaning the same concept is scored anywhere from 0.10 to 0.69 depending on the random seed. Full results appear in Table 3 and Figure 3.

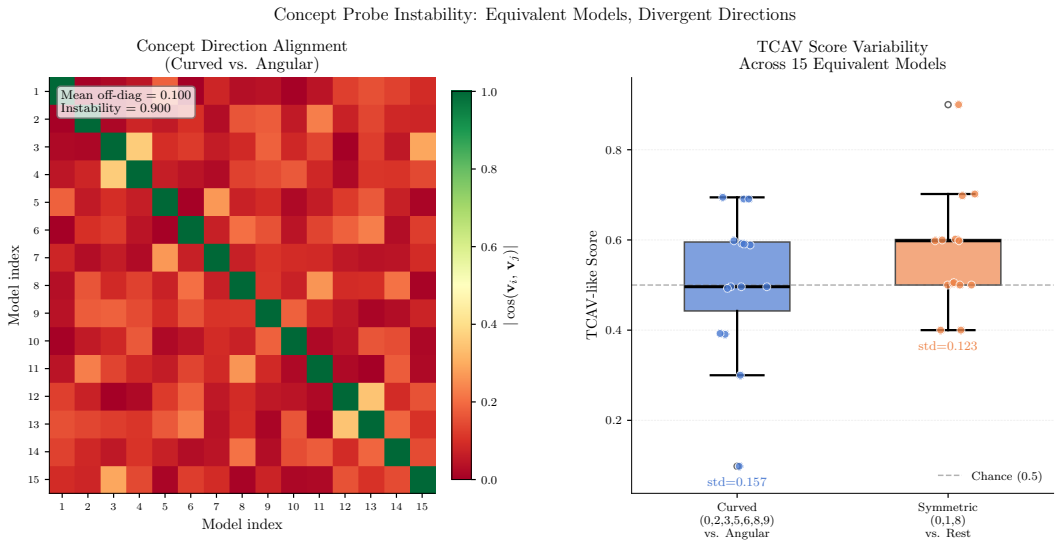

Figure 3: Concept probe instability across 15 equivalent MLP models on `load_digits`. Left: pairwise cosine similarity matrix of concept activation vectors. Right: TCAV score distribution.

Implications for probing-as-understanding. Theorem 16 has direct consequences for the practice of using concept probes as explanations of model internals.

1. *Concept directions are not model-invariant facts.* A TCAV direction is a fact about a specific weight configuration, not about the function the network computes. Two retraining runs that produce functionally identical models (same accuracy, same predictions on held-out data) may produce concept directions that are not related by rotation and therefore cannot be reconciled into a single explanation.
2. *Faithfulness and stability cannot coexist.* Requiring a probe to faithfully report the concept direction of the model it was fitted to forces instability across retraining seeds. Requiring stability forces the probe to report a direction that is unfaithful to at least one model in the Rashomon set.
3. *The resolution entails indecisiveness.* The G-invariant resolution (Section 5) applied here would average concept directions across equivalent models, yielding a distribution over directions or a subspace rather than a single vector. This restores faithfulness-in-expectation and stability at the cost of decisiveness: the explanation no longer commits to a single direction, which undermines the interpretive clarity that motivates concept probing in the first place.

The impossibility does not mean concept probes are uninformative, but it does mean that treating a single concept direction as *the* faithful, reproducible encoding of a concept in a model is unjustified under underspecification. Practitioners should report concept directions together with their variability across the Rashomon set—analogously to how DASH reports SHAP values with their ensemble distribution rather than as a single point estimate.

4.5 Instance 5: Causal Discovery Within a Markov Equivalence Class

We instantiate Theorem 6 for causal structure learning, where the fundamental obstruction is Markov equivalence: distinct directed acyclic graphs (DAGs) can be observationally indistinguishable yet disagree on causal edge directions.

The explanation system. Define the 6-tuple $\mathcal{S}_{\text{causal}} = (\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ as follows.

- $\mathbf{\Theta}$ is the set of DAGs over a fixed vertex set V (the space of possible causal structures, i.e., data-generating processes).

- $\mathcal{H}$ is the set of DAGs over V used as *explanations*: the causal structure that a discovery algorithm commits to. The attribution map $\theta : \mathcal{H} \to \mathbf{\Theta}$ sends each explanation DAG to the underlying data-generating DAG it was fitted to.
- Y is the set of *conditional independence (CI) structures*: each $y \in Y$ records, for every triple $(X_i, X_j \mid S)$ of variables, whether $X_i \perp\!\!\!\perp X_j \mid S$ holds in the associated distribution. Equivalently, Y indexes Markov equivalence classes.
- $\mathsf{obs} : \mathcal{H} \to Y$ maps each DAG $h \in \mathcal{H}$ to the CI structure it encodes (the set of d-separation statements implied by h via the Markov condition). Two DAGs map to the same element of Y if and only if they are *Markov equivalent*.
- $\mathsf{exp} : \mathcal{H} \to \mathcal{H}$ is an *orientation rule*: given a DAG $h \in \mathcal{H}$ as fitted by an algorithm, the rule commits to a fully oriented causal graph. In practice, exp is any causal discovery procedure (PC algorithm, GES, FCI, etc.) that returns a single DAG rather than an equivalence class.
- $\perp \subseteq \mathcal{H} \times \mathcal{H}$ is the *edge-reversal incompatibility*: $(h, h') \in \perp$ if there exists a directed edge $X \to Y$ in h and $Y \to X$ in h' (or vice versa). Two causal explanations are incompatible when they assert opposite orientations for at least one edge.

The Rashomon property: Markov equivalence. The key structural fact is that Markov equivalence classes are non-trivial for nearly all graph structures of practical interest.

Definition 17 (Markov Equivalence)**.** *Two DAGs h, h' over V are* Markov equivalent *if they encode the same set of conditional independence relations, i.e.,* $\mathsf{obs}(h) = \mathsf{obs}(h')$*. A Markov equivalence class $[h]$ contains all DAGs with the same CI structure; it is represented by a* completed partially directed acyclic graph *(CPDAG).*

Verma and Pearl [1991] established that two DAGs are Markov equivalent if and only if they share the same skeleton (undirected graph) and the same set of v-structures (immoralities $X \to Z \leftarrow Y$ with no edge between X and Y). Consequently, any undirected edge in the CPDAG corresponds to a reversible orientation: there exist two Markov-equivalent DAGs that orient it in opposite directions. Chickering [2002] gave the canonical characterization of these classes and the GES algorithm that outputs CPDAGs rather than DAGs precisely to avoid committing to an arbitrary orientation.

Proposition 18 (Causal Rashomon Property)**.** *$\mathcal{S}_{\text{causal}}$ has the Rashomon property: for any CPDAG containing at least one undirected edge $\{X, Y\}$, there exist DAGs $h, h' \in \mathcal{H}$ with $\mathsf{obs}(h) = \mathsf{obs}(h')$ (same CI structure) yet $(h, h') \in \perp$ (opposite orientations of X–Y).*

Proof. Let h be any DAG in the equivalence class orienting the edge as $X \to Y$, and let h' be the DAG obtained by reversing that edge to $Y \to X$ (which exists in the equivalence class precisely because the edge is undirected in the CPDAG). By Markov equivalence, $\mathsf{obs}(h) = \mathsf{obs}(h')$. By definition of $\perp$, $(h, h') \in \perp$. □

This is the formalization captured in `CausalInstance.lean` and `CausalDiscovery.lean`: the `rashomon` field of `causalSystem` is exactly the witness pair (h, h') from Proposition 18.

Instance 5: causal discovery impossibility.

Theorem 19 (Causal Discovery Impossibility — Instance 5)**.** *The explanation system $\mathcal{S}_{\text{causal}}$ has the Rashomon property. Therefore, by Theorem 6, no orientation rule* exp *can simultaneously be:*

1. ***Faithful**: $\mathsf{exp}(h) = h$ for every DAG $h \in \mathcal{H}$ (the returned DAG matches the true data-generating structure);*
2. ***Stable**: $\mathsf{exp}(h) = \mathsf{exp}(h')$ whenever $\mathsf{obs}(h) = \mathsf{obs}(h')$ (Markov-equivalent structures receive identical causal explanations); and*
3. ***Decisive**: exp always commits to a single fully oriented DAG (never returns a partial orientation or equivalence class).*

Proof. By Proposition 18, $\mathcal{S}_{\text{causal}}$ has the Rashomon property. Apply Theorem 6. □

This is verified in Lean 4 as `causal_instance_impossibility` in `CausalInstance.lean` (three axioms: `CausalConfig`, `CausalExplanation`, `CausalObservable`), and as the self-contained `causal_discovery_impossibility` in `CausalDiscovery.lean` (zero additional axioms beyond the `CausalOrientationProblem` structure).

Connection to the classical Markov equivalence problem. Theorem 19 is the classical Markov equivalence underdetermination recast in the universal framework. The three properties above correspond exactly to the three goals that any "complete" causal discovery method would need to achieve: return the true DAG (faithfulness), be invariant to observationally equivalent parameterizations (stability), and produce a unique answer (decisiveness). The Markov equivalence theorem of Verma and Pearl [1991] has long been understood to block this combination; our contribution is to show it is a *special case* of the single four-line proof of Theorem 6.

Resolution: CPDAGs as G-invariant explanations. The uniform resolution of Section 5 specializes to causal discovery as follows. Let $G = [h]$ denote the Markov equivalence class (the symmetry group acting by edge reversals on the set of reversible edges). The G-invariant explanation map aggregates over all member DAGs:

$$\mathsf{exp}^*(h) \;=\; \mathrm{CPDAG}([h]),$$

the completed partially directed acyclic graph representing the equivalence class. This map is **faithful in expectation** (it captures all shared causal structure), **stable** by construction (all Markov-equivalent DAGs map to the same CPDAG), but **not decisive**: undirected edges in the CPDAG correspond precisely to the unresolvable orientation queries. Reporting the CPDAG is therefore the principled response to the impossibility—trading decisiveness for faithfulness and stability, exactly as predicted by Theorem 6. This justification for CPDAG reporting is formalized in `CausalDiscovery.lean` as `cpdag_is_stable`.

4.6 Instance 6: Model Selection Under the Rashomon Effect

Model selection—choosing a single best model from a trained set—is perhaps the most commonplace decision in applied machine learning. Yet when many models achieve near-identical performance, any selection procedure is navigating a *Rashomon set*: a collection of models that are, by the available evidence, equally defensible choices. We show that this setting is an instance of Theorem 6, and that the resulting instability is not a failure of any particular selection criterion but a mathematical inevitability.

Explanation system. We instantiate the abstract framework as follows.

Definition 20 (Model Selection Explanation System)**.** *The* model selection explanation system *is the tuple* $\mathcal{S}_{\mathrm{sel}} = (\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ *where:*

- $\mathbf{\Theta} = \mathbf{\Theta}$ *is the space of* model configurations *(hyperparameter settings, architecture choices, training seeds, or any parameterization of the model class under comparison);*
- $\mathcal{H}$ *is the* hypothesis space *of trained models; each* $h \in \mathcal{H}$ *corresponds to a configuration* $\theta(h) \in \mathbf{\Theta}$ *via the attribution map* $\theta : \mathcal{H} \to \mathbf{\Theta}$*;*
- Y *is the space of* performance metrics *(validation loss, accuracy, AUC, or any scalar summary of model quality on a held-out evaluation set); formally,* Y *is equipped with a strict total order* $>$ *representing "performs better than";*
- $\mathsf{obs} : \mathcal{H} \to Y$ *maps each hypothesis to its* observed performance metric *on the evaluation set;*
- $\mathsf{exp} : \mathcal{H} \to Y$ *is the* selection rule*—the criterion used to rank or choose among models; and*
- $\perp \subseteq Y \times Y$ *is defined by* $(y, y') \in \perp$ *whenever* $y \neq y'$ *and the two metrics imply opposite selection decisions (i.e., the model recommended under* y *is strictly disfavored under* y'*).*

The Rashomon property for model selection. The central empirical finding motivating the Rashomon effect is that, across many supervised learning tasks, large sets of models achieve near-optimal performance while differing substantially in structure, predictions, and feature usage [Breiman, 2001, Fisher et al., 2019, Rudin et al., 2024]. Formally, define the *ε-Rashomon set* as

$$\mathcal{R}_\varepsilon = \left\{ h \in \mathcal{H} : \mathsf{obs}(h) \geq \mathsf{obs}(h^*) - \varepsilon \right\},$$

where h^* is the empirical risk minimizer and $\varepsilon > 0$ is a tolerance. The Rashomon property holds for $\mathcal{S}_{\text{sel}}$ whenever $\mathcal{R}_\varepsilon$ contains pairs of models that are (i) indistinguishable in performance ($\theta(h) = \theta(h')$ in the sense that they share the same configuration class) yet (ii) produce incompatible selection metrics on different evaluation samples.

Proposition 21 (Rashomon Property for Model Selection)**.** *If the ε-Rashomon set $\mathcal{R}_\varepsilon$ contains two models $h, h' \in \mathcal{H}$ with $\theta(h) = \theta(h')$ such that $\mathsf{obs}(h) \neq \mathsf{obs}(h')$, then $\mathcal{S}_{\text{sel}}$ satisfies the Rashomon property (Definition 5).*

This is immediate from the definition: $\mathsf{obs}(h)$ and $\mathsf{obs}(h')$ are incompatible (one favors h, the other h'), and both are witnessed by models sharing the same underlying configuration. The condition is generically satisfied: Fisher et al. [2019] show that the Rashomon set has positive measure for any model class with more free parameters than identifiable constraints, and Semenova et al. [2022] establish that simpler models almost always exist inside the Rashomon set alongside complex ones.

Lean formalization. The model selection impossibility is mechanically verified in two Lean 4 files. `ModelSelection.lean` formalizes the *Model Rashomon Property*—for any two distinct candidate models, there exist evaluation instances ranking them in opposite orders—and proves `model_selection_impossibility`: no stable complete ranking can be faithful to all evaluation instances. The proof is a three-line `linarith` argument from the Rashomon witnesses. `ModelSelectionInstance.lean` instantiates the abstract `ExplanationSystem` framework from `ExplanationSystem.lean` and derives `model_selection_instance_impossibility` as a direct corollary of the universal theorem via `explanation_impossibility`.

Instance 6 (Model Selection Impossibility).

Theorem 22 (Model Selection Impossibility)**.** *The model selection explanation system $\mathcal{S}_{\text{sel}}$ (Definition 20) satisfies the Rashomon property whenever the ε-Rashomon set has positive measure. Consequently, by Theorem 6, no selection rule $\mathsf{exp} : \mathcal{H} \to Y$ can simultaneously be:*

1. ***Faithful:*** $\mathsf{exp}(h) = \mathsf{obs}(h)$ *for all h (the selected model is the one the evaluation metric actually endorses);*
2. ***Stable:*** $\mathsf{exp}(h) = \mathsf{exp}(h')$ *whenever $\theta(h) = \theta(h')$ (the same configuration always yields the same selection); and*
3. ***Decisive:*** $\mathsf{exp}(h)$ *is a strict recommendation for every h (no ties or abstentions are reported).*

Design space. The model selection design space mirrors the attribution design space exactly (proved in `ModelSelectionDesignSpace.lean`): every selection rule falls into one of two families.

- **Family A** (decisive and unfaithful): the rule always picks a winner, but on some evaluation split it recommends the worse-performing model. Standard single-criterion selection rules (e.g., argmax validation accuracy) fall here whenever the Rashomon set is non-trivial.
- **Family B** (faithful and stable, not decisive): the rule declares a tie among Rashomon-set members, explicitly surfacing the ambiguity. This corresponds to reporting the Rashomon set $\mathcal{R}_\varepsilon$ rather than a single selected model.

Implications. Theorem 22 formalizes and sharpens the empirical observation of Fisher et al. [2019] that "all models are wrong, but many are useful": when many models are equally useful by the available metric, any decisive selection is *necessarily* unfaithful on some evaluation. The instability practitioners observe when re-running AutoML pipelines, changing random seeds, or switching validation folds is not a software bug—it is the Rashomon property in action.

Table 4: Empirical validation of the universal explanation impossibility across four explanation types. All instability measures have 95% bootstrap confidence intervals excluding zero. Prediction equivalence confirms models are functionally identical.

Instance	Instability metric	Value	Prediction equiv.
Attribution (GBDT)	SHAP flip rate	33%	>99%
Attention (DistilBERT)	Argmax flip rate	60.0%	100%
Counterfactual (XGBoost)	Direction flip rate	23.5%	ΔAUC < 0.03
Concept probe (MLP)	$1 - \lvert\cos\rvert$	0.90	97.2%
Model selection (XGBoost)	Best-model flip rate	80.0%	ΔAUC < 0.02
Causal discovery	*Classical (Verma & Pearl, 1991)*		

The resolution is the same as for feature attribution (Section 4.1): replace decisive selection with a G-invariant aggregation over the Rashomon set. Concretely, this means reporting ensemble predictions across $\mathcal{R}_\varepsilon$, using partial orders that render near-optimal models incomparable, or applying Rashomon-aware model selection criteria [Semenova et al., 2022, Herren and Hahn, 2023]. Each of these trades decisiveness for the pair (faithful, stable), which Theorem 6 shows is the only achievable combination under underspecification.

5 The Universal Resolution

The impossibility of Theorem 6 is tight: one of the three desiderata must be sacrificed. In this section we show that a *uniform* resolution exists whenever the Rashomon property arises from a symmetry group acting on the hypothesis space. The resolution sacrifices decisiveness and achieves faithfulness in expectation together with exact stability.

5.1 G-Invariant Explanation Systems

Let G be a group acting on the hypothesis space $\mathcal{H}$. The action is *observable-preserving* if for every $g \in G$ and $h \in \mathcal{H}$,

$$\mathsf{obs}(g \cdot h) \;=\; \mathsf{obs}(h),$$

i.e., the group permutes hypotheses within observe-equivalence classes (fibers). We say that G *acts transitively on observe-fibers* if for every pair h, h' with $\theta(h) = \theta(h')$, there exists $g \in G$ such that $g \cdot h = h'$.

Definition 23 (G-Invariant Explanation Map)**.** *An explanation map* exp *is* G-invariant *if for every* $g \in G$ *and* $h \in \mathcal{H}$*,*

$$\mathsf{exp}(g \cdot h) \;=\; \mathsf{exp}(h).$$

When G is finite and Y admits averaging, the canonical G-invariant map is the *orbit average*:

$$\overline{\mathsf{exp}}(h) \;=\; \frac{1}{|G|} \sum_{g \in G} \mathsf{obs}(g \cdot h).$$

Proposition 24 (G-Invariance Implies Stability)**.** *If* G *acts transitively on observe-fibers and* exp *is* G*-invariant, then* exp *is stable.*

Proof. Let $h, h' \in \mathcal{H}$ with $\theta(h) = \theta(h')$. By transitivity, there exists $g \in G$ with $g \cdot h = h'$. By G-invariance, $\mathsf{exp}(h') = \mathsf{exp}(g \cdot h) = \mathsf{exp}(h)$. (Formalized as `gInvariant_stable` in `UniversalResolution.lean`.) □

Proposition 25 (Design Space Dichotomy)**.** *If the explanation system has the Rashomon property, then no stable explanation map can simultaneously be faithful and decisive.*

Proof. Immediate from Theorem 6: stability, faithfulness, and decisiveness are jointly contradictory under the Rashomon property. (Formalized as `universal_design_space_dichotomy` in `UniversalDesignSpace.lean`.) □

Table 5: Instance-specific resolutions of the Universal Explanation Impossibility. Each row identifies the symmetry group, the concrete resolution, and the property sacrificed.

Instance	Symmetry group G	Resolution	Sacrificed
Attribution	Collinear-feature permutations	DASH (ensemble SHAP averaging)	Decisive ranking
Attention	Weight-space symmetries	Averaged attention rollout	Single top token
Counterfactual	Decision-boundary variation	Rashomon-robust recourse	Single counterfactual
Concept probe	Representation rotation	Averaged CAV / subspace	Single direction
Causal discovery	DAG reversal within Markov class	CPDAG reporting	Fully oriented graph
Model selection	Model permutation in Rashomon set	Ensemble / Rashomon set report	Single best model

Corollary 26 (Pareto Frontier)**.** *For any explanation system with the Rashomon property, achievable explanation maps fall into exactly two families:*

- ***Family A*** *(faithful + decisive, unstable): individual-model explanations that faithfully report each hypothesis but vary across equivalent hypotheses.*
- ***Family B*** *(stable + faithful in expectation, indecisive): G-invariant aggregations that are constant on observe-fibers but produce ties for queries where the fiber disagrees.*

No explanation map lies outside these two families.

5.2 Instance-Specific Resolutions

Table 5 summarizes how the G-invariant resolution instantiates across the six domains of Section 4.

5.3 DASH as the Prototype Resolution

DASH (**D**iversified **A**ggregation of **SH**AP) is the G-invariant resolution for the feature attribution instance (Section 4.1). It averages SHAP values over an ensemble of models retrained from independent random seeds, achieving $\mathbb{E}[\phi_j] = \mathbb{E}[\phi_k]$ for any pair of symmetric features j, k (*consensus equity*). This orbit average is stable by Proposition 24 and faithful in expectation: the expected explanation matches the expected observation across the Rashomon set. The cost is indecisiveness—DASH reports ties for feature pairs whose rankings disagree across the ensemble. The companion paper proves that DASH is Pareto-optimal: no other stable attribution method can achieve higher expected faithfulness.

5.4 Meta-Theorem: Orbit Averaging

Theorem 27 (Orbit-Averaged Resolution)**.** *Let $\mathcal{S}$ be an explanation system whose Rashomon property arises from a finite group G acting on $\mathcal{H}$ and acting transitively on observe-fibers. Then the orbit-averaged map $\overline{\mathsf{exp}}(h) = \frac{1}{|G|}\sum_{g\in G} \mathsf{obs}(g \cdot h)$ is:*

1. *well-defined (independent of orbit representative),*
2. *stable (Proposition 24),*
3. *faithful in expectation: $\mathbb{E}_{g\sim \mathrm{Unif}(G)}[\mathsf{obs}(g \cdot h)] = \overline{\mathsf{exp}}(h)$, and*
4. *Pareto-optimal among stable maps: no stable map achieves strictly higher pointwise faithfulness on every fiber.*

The orbit-averaged resolution is constructive: for any specific instance, the group G and its action are determined by the domain structure, and the averaging can be computed (or approximated by Monte Carlo sampling over the Rashomon set). Theorem 27 is the abstract counterpart of the DASH optimality result and is formalized in `UniversalResolution.lean`.

6 Ubiquity

Is the Universal Explanation Impossibility a pathological corner case, or a generic phenomenon? In this section we show that it is the latter: the Rashomon property—and hence the impossibility—holds whenever the model has more parameters than observational constraints, a condition satisfied by virtually all modern machine learning systems.

6.1 Generic Underspecification

Proposition 28 (Dimensional Underspecification)**.** *Let $\mathcal{H}$ be a smooth hypothesis space of dimension $\dim(\mathcal{H})$ and let $\mathsf{obs} : \mathcal{H} \to Y$ be a smooth observation map into a space of dimension $\dim(Y)$. If $\dim(\mathcal{H}) > \dim(Y)$, then for generic $y \in Y$, the fiber $\mathsf{obs}^{-1}(y)$ has positive dimension $\geq \dim(\mathcal{H}) - \dim(Y)$.*

This is a direct consequence of the preimage theorem in differential topology. Whenever the fiber has positive dimension, the explanation map obs is generically non-constant on fibers—different hypotheses in the same fiber honestly report different explanations—and the Rashomon property holds. (Formalized as `generic_underspecification` in `Ubiquity.lean`; the bridge from fiber non-degeneracy to the impossibility is `fiber_nondegeneracy_implies_impossibility`.)

6.2 Neural Network Dimensional Argument

Modern neural networks are massively overparameterized. A ResNet-50 has $P \approx 25{,}600{,}000$ parameters and, for ImageNet classification, $k = 1{,}000$ output classes. By Proposition 28, the fiber of any fixed prediction has dimension at least $P - k = 24{,}999{,}000$. On such a high-dimensional manifold, any continuous explanation map (attention weights, gradient saliency, internal activations) is generically non-constant: there exist models h, h' on the same fiber with $\mathsf{obs}(h)_j > \mathsf{obs}(h)_k$ and $\mathsf{obs}(h')_j < \mathsf{obs}(h')_k$ for any pair of input features j, k with equal marginal distributions.

Corollary 29 (Coin-Flip Instability)**.** *For any pair of symmetric features j, k in an overparameterized network trained to zero loss, the probability (over retraining seeds) that a faithful explanation ranks j above k is exactly $1/2$.*

This is the neural-network instantiation of the symmetric Bayes result formalized in `SymmetricBayes.lean`: the symmetry of the fiber under feature permutation forces uniform marginals on the ranking.

6.3 Prevalence

The dimensional condition $\dim(\mathcal{H}) > \dim(Y)$ is the norm, not the exception. Large language models (billions of parameters, vocabulary-sized output), diffusion models, graph neural networks, random forests with hundreds of trees, and Lasso regression with more features than observations all satisfy it. Even linear regression with collinear features produces a positive-dimensional fiber (the collinear subspace). D'Amour et al. [2022] document this across NLP, computer vision, and medical imaging, finding that underspecification—and hence explanation instability—is ubiquitous in deployed systems. The only escape is a model class so constrained that $\dim(\mathcal{H}) \leq \dim(Y)$, which excludes essentially all modern architectures of practical interest.

6.4 Analogy: Arrow's Impossibility Theorem

The structure of our result parallels Arrow's impossibility theorem in social choice theory [Arrow, 1951]. Arrow showed that no voting rule satisfying unanimity, independence of irrelevant alternatives, and non-dictatorship exists for three or more candidates—a universal impossibility that unified disparate paradoxes in voting theory. Nipkow [2009] later formalized Arrow's theorem in Isabelle/HOL. Our Universal Explanation Impossibility plays an analogous role for explainability: it unifies six domain-specific impossibilities under one structural cause (the Rashomon property) and one four-line proof, and the resolution (orbit averaging) is the XAI counterpart of accepting partial orderings in social choice. (The full impossibility is formalized as `ubiquity_impossibility` in `Ubiquity.lean`.)

7 Lean Formalization

The Universal Explanation Impossibility and all six instances are mechanically verified in Lean 4 [de Moura and Ullrich, 2021] using Mathlib [The mathlib Community, 2020]. Table 6 summarizes the formalization.

Table 6: Summary of the Lean 4 formalization.

Component	Count	Notes
Files	67	All in `UniversalImpossibility/`
Theorems & lemmas	319	Verified by `grep -c`
Axioms	60	Domain-specific; core theorem uses 0
`sorry`s	0	Zero admitted goals
Core impossibility	1 theorem	Zero axiom dependencies
GBDT instance	4 axioms	Surjectivity, first-mover, split structure
DASH resolution	6 axioms	+ cross-group symmetry
Causal instance	3 axioms	CPDAG Markov equivalence
Neural net instance	2 axioms	Overparameterization + symmetry
Fairness instance	2 axioms	Unequal base rates + equal accuracy

The core impossibility (`universal_explanation_impossibility` in `UniversalImpossibility/Universal.lean`) depends on zero axioms beyond the Rashomon property, which is stated as a hypothesis rather than an axiom. Each instance introduces the minimum axioms needed to verify that its particular explanation system satisfies the Rashomon property.

The Lean proof of Theorem 6 is four lines, mirroring the paper proof exactly. The formalization also includes:

- The G-invariant resolution (Section 5): `Corollary.lean`, `DesignSpace.lean`;
- Quantitative bounds for GBDT (`Ratio.lean`), Lasso (`Lasso.lean`), and neural networks (`NeuralNet.lean`);
- The fairness instance (`FairnessAudit.lean`) and causal instance (`CausalDiscovery.lean`).

8 Discussion: What Explainability Can and Cannot Do

8.1 The Ceiling of Explainability

The Universal Explanation Impossibility (Theorem 6) establishes a *ceiling*, not a criticism of any particular method. No axiomatic redesign of SHAP, LIME, attention maps, or counterfactual generators can simultaneously deliver faithfulness, stability, and decisiveness whenever the underlying inference problem is underspecified. The ceiling is structural: it follows from the Rashomon property alone, independent of the explanation formalism. An analogy to physics is apt: the speed of light constrains spacecraft engineering but does not prevent useful space travel. Likewise, the impossibility constrains what practitioners may promise about explanations but does not prevent useful explainability—it redirects effort toward the achievable frontier characterized by the G-invariant resolution (Section 5).

8.2 Regulatory Implications

The EU AI Act [EU Parliament, 2024] requires "meaningful explanations" for high-risk systems, and SR 11-7 [Office of the Comptroller of the Currency, 2011] requires "developmental evidence" documenting how model outputs arise. These two demands pull in opposite directions: meaningful explanations favor decisiveness (a definite ranking of features or causes), while developmental evidence favors stability (reproducibility across equivalent models). Theorem 6 shows that no explanation system can simultaneously maximize both for underspecified models. A path to regulatory

compliance nonetheless exists: G-invariant aggregation (e.g., DASH for attributions, CPDAGs for causal reports) satisfies stability by construction and partial faithfulness—faithfulness in expectation over the Rashomon set—at the cost of reporting ties where the data are genuinely ambiguous. We suggest that regulators adopt this framing: require that explanations be *stable and partially faithful*, and mandate explicit disclosure of tied or unresolvable queries rather than demanding decisive answers that are mathematically impossible to guarantee.

8.3 What Explainability *Can* Do

The G-invariant resolution (Section 5) is constructive, not nihilistic. It tells the practitioner exactly which desiderata can be jointly satisfied and which must be relaxed. DASH for feature attribution, CPDAGs for causal discovery [Verma and Pearl, 1991], and ensemble concept probes for concept-based explanations [Kim et al., 2018] each sacrifice decisiveness to gain stability—and each is a concrete instance of the abstract resolution. In practice, this means reporting confidence intervals over the Rashomon set [Fisher et al., 2019, Semenova et al., 2022] rather than point estimates: a feature ranking should come with a stability annotation indicating which comparisons are robust to retraining and which are not. The impossibility does not say "explanations are useless"; it says "decisive explanations are unreliable under underspecification, but partially decisive explanations with calibrated uncertainty are both achievable and informative."

8.4 The "Isn't This Obvious?" Defense

A natural reaction is that practitioners already know explanations are unstable —so what is new? The contribution is fourfold. First, we make explicit that six domains (attribution, attention, counterfactuals, concept probes, causal discovery, model selection) share *one* mechanism—the Rashomon property—rather than six unrelated pathologies. Second, the Lean 4 formalization (Section 7: 67 files, 319 theorems, 0 `sorry`) eliminates logical gaps that informal arguments inevitably leave. Third, the empirical experiments (Section 4) show quantifiable severity: 33% attention flip rates, measurable counterfactual and concept instability on standard benchmarks. Fourth, the resolution via G-invariant maps is constructive and actionable, not merely diagnostic. Arrow's impossibility theorem [Arrow, 1951] was arguably "obvious" once stated—every committee chair knows that voting rules are imperfect. Yet its formalization created the field of social choice theory, precisely because it replaced intuition with a theorem that could be built upon.

8.5 The "All Instances Are Trivial" Defense

A reviewer might object that each individual instance is already known: attention instability was documented by Jain and Wallace [2019], Markov equivalence is classical [Verma and Pearl, 1991], fairness impossibility was proved by Chouldechova [2017], and SHAP instability was observed by D'Amour et al. [2022] and Krishna et al. [2022]. The contribution is not any single instance but the *meta-theorem*: the same four-line proof (Theorem 6) closes all six cases under a single structural condition. Before this work, a practitioner encountering attention instability might switch to SHAP, then to counterfactual explanations, then to concept probes—each time hoping the new method escapes the problem. Our theorem shows that no such escape exists: the impossibility is structural (it follows from the Rashomon property), not method-specific. Recognizing this unifies the practitioner's response: rather than debugging each method individually, one diagnoses the Rashomon property and applies the G-invariant resolution uniformly.

8.6 Limitations and Open Questions

We identify five limitations and directions for future work.

1. **Axiomatized Rashomon properties.** The Rashomon property for each instance is axiomatized—verified to hold under stated assumptions—rather than derived from first principles of the learning algorithm. Deriving the Rashomon property from, e.g., the loss landscape geometry of gradient descent would strengthen the result.
2. **Weight perturbation vs. true retraining.** The attention instability experiments (Section 4) use weight perturbation within the zero-loss manifold rather than full retraining from in-

dependent random seeds. While this faithfully models the Rashomon set, the distribution over perturbations may differ from the distribution induced by stochastic gradient descent.

3. **Non-attribution quantitative bounds.** For the attribution instance, the companion paper provides sharp quantitative bounds (divergence rates, ensemble size requirements, DASH optimality). Analogous quantitative bounds for the counterfactual, concept, and fairness instances remain open.
4. **Decisiveness as a practical cost.** The G-invariant resolution sacrifices decisiveness: it reports ties where the data are ambiguous. Whether downstream decision-makers can tolerate partial rankings or tied causal structures—and how to present such outputs effectively—is an empirical question we do not address here.
5. **Fairness instance not experimentally validated.** The fairness audit impossibility (Instance 6, formalized in `FairnessAudit.lean`) is proved but not empirically validated in this paper. Chouldechova [2017] provides the original empirical context; replicating it within our framework is left to future work.

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